# Scientific Research Methodology
# 1. Introduction

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# Discussion

- What is research?
- How is research conducted?
- How are research results checked and validated?
- How are research results shared?

# What is research?

**Definition**

The process of extending the boundaries of current knowledge in a particular field.

Other equivalent definitions?

**Examples:**

- proving a new theorem in mathematics
- proposing a new, more efficient algorithm to solve a problem in computer science
- extending a known result to a new class of problems

**Question**

What is the difference between **research** and **solving a problem**?

# Research vs. problem solving

**Problem solving**

⋆ using specific, established knowledge to find the answer to a given problem

⋆ it is known *a priori* that the answer can be found by combining known definitions and properties; the truth of the statement is not in question

⋆ **Objective**: consolidating and applying the concepts and ideas learned

**Scientific research**

⋆ finding the answer to an **open** problem: it is not known *a priori* whether the problem can be solved

⋆ new concepts and methods may be needed to obtain research results

# Examples

**Problems:**

- Prove that there are infinitely many prime numbers
- Prove that $\sum_{p \text{ prime}} \frac{1}{p} = +\infty$

**Research questions:**

- **Goldbach's conjecture:** Prove that every even integer greater than 2 can be written as the sum of two prime numbers.
- **Mersenne primes:** Are there infinitely many prime numbers of the form $2^n - 1$?

**Large prime numbers**

The largest known prime number (as of October 2025) is $2^{136279841} - 1$

# University experiences that encourage research

⋆ writing a master's dissertation: exploring a field in depth, with possible original extensions
⋆ participating in student conferences and presenting results
⋆ attending scientific seminars

**The doctoral thesis:**

- represents a first concrete step into research
- obtaining significant new results is a necessary condition for earning a doctorate in a given field

# Aims of this course

- Describe the fundamental stages of scientific research
  - literature review, critical thinking, and evaluating content found online across different publication platforms
  - practical ideas for developing effective research strategies
  - publication venues, peer review, bibliometrics, and research impact
- Case studies: analysing scientific papers
  - identifying the basic elements of a paper
  - reading scientific papers efficiently
  - identifying potential problems and red flags in scientific papers

# Assessment

- Preparing and presenting a project (reading and analysing a given scientific paper): 80%
- Active participation: 20%